

From GPU-Accelerated Simulation to Scalable and Generalizable Real Robot Policy Learning

November 12, 2026 • Austin, Texas, USA • Half-Day Workshop
• <https://corl2026-realworld-robot-learning.github.io>

“Simulation is the most scalable training ground for embodied AI — from rigorous benchmarks and diagnostic evaluation protocols, to reliable transfer to real-world deployment.”

WORKSHOP OVERVIEW



SCAN FOR
FULL PROGRAM

Simulation has become the engine of modern embodied AI. Physics-grounded environments like **BEHAVIOR** deliver the scale, safety and reproducibility that physical data collection cannot. Diagnostic frameworks like **RoboEval** instrument every task with stage-level metrics that reveal not just whether a policy succeeds, but precisely how and why it fails. **RoboChess** puts simulation-trained policies to the ultimate test by requiring direct transfer to physical robots. The frontier question is not whether to train in simulation, but **how to ensure that capabilities learned at scale carry over robustly to the physical world.**

CALL FOR PAPERS — TOPICS OF INTEREST

- Simulation environments and benchmarks for scalable robot learning and evaluation
- Diagnostic and fine-grained evaluation: capability metrics, failure analysis, robustness, reproducibility
- Sim-to-real transfer and validation: domain shift, adaptation, simulated-vs-real performance studies
- Challenging robot capabilities: long-horizon, dexterous, bimanual and generalist robot learning

INVITED SPEAKERS



Yashraj Narang
NVIDIA



Guannan Qu
Carnegie Mellon University



Yilun Du
Harvard University



Chen Tang
UC Los Angeles

KEY DATES

Submission Open	Aug 26, 2026
Paper Deadline	Oct 9, 2026
Notification	Oct 23, 2026
Workshop Day	Nov 12, 2026

Embodied AI Benchmark & Sim-to-Real Challenge Suite

Three tracks · independent leaderboards · shared evaluation protocol

TRACK 1 BEHAVIOR

Long-Horizon Household Challenge

1,000 household activities across 50 scenes and 10,000 objects on the Isaac Sim-based BEHAVIOR simulator — reasoning, navigation and dexterous bimanual manipulation over long horizons.

behavior.stanford.edu

TRACK 2 RoboEval

Diagnostic Bimanual Challenge

Stage-level instrumentation beyond binary success — outcome, efficiency, bimanual coordination and safety. 3,000+ VR-teleoperated expert demonstrations.

robo-eval.github.io

TRACK 3 RoboChess

Sim-to-Real Manipulation Challenge

Policies trained in GPU-accelerated simulation must transfer to physical hardware — perceive a real chessboard, plan long-horizon moves, execute across embodiments.

sites.google.com/view/corl-workshop-robochess

ORGANIZING COMMITTEE



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Call for Papers

- Format:** short papers, up to 5 pages excluding references
- Review:** double-blind on OpenReview
- Presentation:** spotlights & posters, interactive discussion
- Welcome:** methods, benchmarks, challenge reports, empirical studies, negative results, position papers

SUBMIT ON OPENREVIEW

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Contact zhenzhenl@nvidia.com

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